



[BIT703 Web Technologies](#) / [Contents](#) / Topic 10: Server-side web application development with procedural and object-oriented database programming / PHP OOP

Introduction to PHP OOP 35m

What is OOP?

OOP stands for object-oriented programming. It is a programming style that is based on the concept of objects and classes. It is about combining a group of related variables and functions into an object; the object will then contain both the variables and functions.

Variables are referred to as properties or attributes and functions are referred to as methods.

The simplest way to explain OOP is to think about a car. A car has a model name, a colour and an engine type – these are the attributes of the car. A Car object can then be created with the model name, colour and engine as the attributes. Additionally, code can be added to the car object to add functions (called methods) to manipulate the car. For example, you could add `start_car()` to simulate turning the engine on and `brake()` to simulate slowing the car down.

OOP is not a programming language, but it is a programming paradigm that is added to PHP5 to help build complex and reusable code 'objects' for web applications. In the previous topic, you learned that PHP is a scripting language that is used on the server side. The style of PHP coding you learned in the previous topic is referred to as procedural programming. Procedural programming is about writing procedures or functions that perform operations on the data, while OOP is about creating objects that contain both data and functions.

[Skip to main content](#)

The four principles of OOP

OOP has four basic principles: encapsulation, abstraction, inheritance and polymorphism. Understanding these principles will help you understand OOP as a programming style.

- **Encapsulation** refers to encapsulating (enclosing) all the data and methods together to form an object. This also includes information hiding to prevent direct access to the data within the class.
- **Data abstraction** refers to any representation of data in which the implementation details are hidden (abstracted).
- **Inheritance** refers to when a class is defined by inheriting the existing function of a parent class. Here, a child class will inherit all or a few methods and variables of a parent class.
- **Polymorphism** refers to when a function/method of the same name can be used for different purposes. For example, the function name will remain the same, but it will take a different number of arguments and can do different tasks.

Watch *Why object-oriented programming* (Codecademy, 2019) (11:12)^{fnref:869524} to learn more about the four concepts of OOP (abstraction, encapsulation, inheritance and polymorphism). This video provides easy-to-understand explanations and examples of each of the four concepts of OOP. It also explains what OOP is and how it differs from procedural programming.



[Skip to main content](#)

Why Object Oriented Programming

Codecademy



Watch on

SHOW TRANSCRIPT



☒ Procedural programming vs OOP

Non-Assessed

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Drag and drop statements into the correct categories: 'Procedural programming' or 'Object-oriented programming (OOP)' based on their descriptions. Once all cards have been placed, click 'Submit' to check your answers.

Aspect	Procedural Programming
Definition	
Data handling	
Application size	
Ease of adding data/methods	

❖ Does not support data hiding

❖ Uses objects and classes to model real-world environments

❖ Designed for complex and large web applications

❖ Difficult to add new data and functions

❖ Supports data hiding within objects and classes

❖ Designed for medium-sized web applications

[Skip to main content](#)

❖ Code written as procedures or functions executed step-by-step

❖ Easier to add new data and functions

Submit



☒ Four concepts of OOP

Non-Assessed

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Below you will find a car example, which explains OOP using a real-world example.

Car example

The simplest way to explain OOP is to think about a car. A car has a model name, a colour and an engine type – these are the attributes of the car. A car object can then be created with the model name, colour and engine as the attributes. Additionally, code can be added to the car object to add functions (called methods) to manipulate the car. For example, you could add `start_car()` to simulate turning the engine on and `brake()` to simulate slowing the car down.

The statements below relate to this car example and explain one of the four OOP principles. Match the statements to the correct OOP principle that the statement relates to.

Encapsulation		
Inheritance		
Abstraction		
Polymorphism		

[Skip to main content](#)



The attributes of the car (model, engine type and colour) and the methods (start_car() and brake()) are grouped together into a Car object.



When we simplify the car object to create a vehicle object with just wheels, we can then create vehicles such as trucks and buses because they share similar attributes.



The method start() will have some code attached to it to explain how the code is to simulate turning the engine on. However, the car object calling on this method does not need to know how the code works or what part of the code makes it work; all the object needs to know is that the engine turns on.



The start() method used in the car object can be applied to a Motorbike object. However, this method will be implemented differently for the Motorbike object compared to the Car object.

Submit



Dig deeper

To learn more about the basics of PHP OOP, refer to these resources:

[Skip to main content](#)

- *PHP - What is OOP?* (W3Schools, n.d.)^{fnref:8becfb}
- *Object Oriented Programming in PHP* (Tutorials Point, n.d.)^{fnref:59ebed}
- *Learn object oriented PHP – 3 hour PHP OOP course* (Clarke, 2022) (3:00:41)^{fnref:08005e}

LinkedIn Learning

If you have a New Zealand library card, you can access LinkedIn Learning video resources for free. Instructions for how to log in to LinkedIn Learning using your library card can be found on LinkedIn Learning for Library – Patron FAQ.

- *PHP: Object-oriented programming* (Skoglund, 2022) (3:53:00)^{fnref:2fb999}



Thoughts on this page?

